

healthy individuals, can result in severe pneumonia in people with AIDS. Such opportunistic diseases, as well as nerve damage and body wasting, are the primary causes of death from AIDS, rather than HIV itself.

Transmission of HIV requires the transfer of virus particles or infected cells from person to person via body fluids such as semen, blood, or breast milk. Unprotected sex (that is, without using a condom) and transmission via HIV-contaminated needles (often among intravenous drug users) cause the vast majority of HIV infections. The virus can enter the body through mucosal linings of the vagina, vulva, penis, or rectum during intercourse or via the mouth during oral sex. People infected with HIV can transmit the disease in the first few weeks of infection, *before* they express HIV-specific antibodies that can be detected in a blood test. Although no cure has been found for HIV infection, drugs that can significantly slow HIV replication and the progression to AIDS are available.

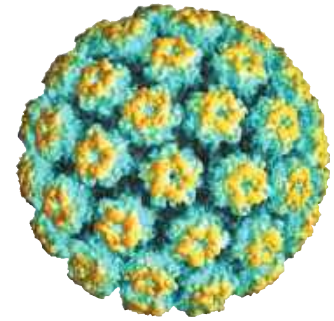
➔ **Mastering Biology Animation: HIV Reproductive Cycle**

## Cancer and Immunity

When adaptive immunity is inactivated, the frequency of certain cancers increases dramatically. For example, the risk of developing Kaposi's sarcoma is 20,000 times greater for patients with untreated AIDS than for people who are healthy. This observation was initially puzzling. If the immune system recognizes only nonself, it should fail to recognize the uncontrolled growth of self cells that is the hallmark of cancer. It turns out, however, that viruses are involved in about 15–20% of all human cancers. Because the immune system can recognize viral proteins as foreign, it can act as a defense against viruses that can cause cancer and against cancer cells that harbor viruses.

Scientists have identified six viruses that can cause cancer in humans. The Kaposi's sarcoma herpesvirus is one such virus (see Table 43.1). Hepatitis B virus, which can trigger liver cancer, is another. A vaccine introduced in 1986 for hepatitis B virus was the first vaccine shown to help prevent a specific human cancer. Rapid progress on developing vaccines for virus-induced cancers continues. In 2006, the release of a vaccine specific for human papillomavirus (HPV) marked a major victory against cervical cancer, as well as oral cancers that are becoming increasingly common among men. The computer graphic image of an HPV particle in **Figure 43.29** illustrates the abundant copies of the capsid protein (yellow) that is used as the antigen in vaccination.

▼ **Figure 43.29** Human papillomavirus.



➔ **Mastering Biology**  
**Interview with Harald zur Hausen:**  
**Discovering the cause of cervical cancer**



### CONCEPT CHECK 43.4

1. In the condition known as myasthenia gravis, antibodies bind to and block certain receptors on muscle cells, preventing muscle contraction. What type of disorder is myasthenia gravis?
2. People with herpes simplex type 1 viruses often get mouth sores when they have a cold or similar infection. How might this location benefit the virus?
3. **WHAT IF?** How would a macrophage deficiency likely affect a person's innate and adaptive defenses?

*For suggested answers, see Appendix A.*

# 43 Chapter Review



➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

## SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocab Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to [goo.gl/zkzj9t](http://goo.gl/zkzj9t).

### CONCEPT 43.1

**In innate immunity, recognition and response rely on traits common to groups of pathogens**  
(pp. 953–957)

- **Innate immunity** is mediated by physical and chemical barriers as well as cell-based defenses. Recognition proteins in innate immunity are specific for broad classes of **pathogens**. Pathogens that penetrate barrier defenses are ingested by phagocytic cells,

which in vertebrates include **macrophages** and **dendritic cells**. Additional cellular defenses include **natural killer cells**, which can induce the death of virus-infected cells. **Complement system** proteins, **interferons**, and other antimicrobial peptides also act against pathogens. In the **inflammatory response**, **histamine** and other chemicals that are released at the injury site promote changes in blood vessels that enhance immune cell access and action.

- Pathogens sometimes evade innate immune defenses. For example, some bacteria have an outer capsule that prevents recognition, while others are resistant to breakdown within lysosomes.

❓ *In what ways does innate immunity protect the mammalian digestive tract?*